

Link to code: <https://github.com/luca-miniati/ds4400/blob/main/hw1/hw1.ipynb>

Solution 1

1. $E[X] = \frac{1}{50}(3 \cdot 25 + 10 \cdot 15 + 5 \cdot 10) = \frac{11}{2}$
2. $E[X^2] = \frac{1}{50}(3^2 \cdot 25 + 10^2 \cdot 15 + 5^2 \cdot 10) = \frac{79}{2}$
 $(E[X])^2 = \frac{121}{4}$
 $\text{Var}(X) = \frac{79}{2} - \frac{121}{4} = \frac{37}{4}$
3. $E[X] = \frac{1}{47}(3 \cdot 23 + 10 \cdot 15 + 5 \cdot 9) = \frac{264}{47}$
4. $E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X] = \text{Var}(X) + (E[X])^2 - E[X] = 2 + 10^2 - 10 = 92$

Solution 2

Let F be the event that an email has "free money", and let S be the event that an email is spam.

1. $P(F) = P(F|S)P(S) + P(F|\neg S)P(\neg S)$

$$P(F) = \frac{1}{10} \cdot \frac{8}{10} + \frac{1}{100} \cdot \frac{2}{10}$$

$$P(F) = \frac{41}{500}$$

2. $P(S|F) = \frac{P(F|S)P(S)}{P(F|S)P(S) + P(F|\neg S)P(\neg S)}$

$$P(S|F) = \frac{\frac{1}{10} \cdot \frac{8}{10}}{\frac{1}{10} \cdot \frac{8}{10} + \frac{1}{100} \cdot \frac{2}{10}}$$

$$P(S|F) = \frac{40}{41}$$

3. $P(\neg S|\neg F) = \frac{P(\neg F|\neg S)P(\neg S)}{P(\neg F|\neg S)P(\neg S) + P(\neg F|S)P(S)}$

$$P(\neg S|\neg F) = \frac{\frac{2}{10} \cdot \frac{99}{100}}{\frac{2}{10} \cdot \frac{99}{100} + \frac{8}{10} \cdot \frac{9}{10}}$$

$$P(\neg S|\neg F) = \frac{11}{51}$$

Solution 3

1. Row-reduce D_1 and D_2 :

$$D_1 = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 2 & 4 & -2 \\ -1 & 0 & -3 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$D_2 = \begin{pmatrix} 1 & -1 & 0 & 2 \\ -1 & 0 & 1 & -1 \\ 0 & -2 & 0 & 1 \\ 3 & 0 & -1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

So the columns of D_1 are linearly dependent, and the columns of D_2 are linearly independent.

2. The maximum number of columns that are linearly independent is 3 for D_1 and 4 for D_2 .
3. The rank is 3 for D_1 and 4 for D_2 , using the RREF's found in (1)
4. The sizes only work for $D_1\theta$ and $\theta^T D_1$:

$$D_1\theta = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 2 & 4 & -2 \\ -1 & 0 & -3 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

$$\theta^T D_1 = (-1 \ 0 \ 1 \ 2) \cdot \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 2 & 4 & -2 \\ -1 & 0 & -3 & 1 \\ 0 & -1 & -1 & 0 \end{pmatrix} = (-2 \ -2 \ -7 \ 1)$$

Solution 4

Let Anton = 1, Geraldine = 2, James = 3, Lauren = 4.

Let English = 1, French = 2, German = 3, Italian = 4, Spanish = 5.

$$1. \ A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$$AA^T = \begin{pmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 2 & 2 \\ 0 & 2 & 3 & 3 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

The (i, j) -th entry in AA^T is the number of shared languages between person i and person j . When $i = j$, the (i, j) -th entry counts the number of languages person i speaks.

$$A^T A = \begin{pmatrix} 3 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 1 & 1 \\ 3 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 & 2 \end{pmatrix}$$

The (i, j) -th entry in $A^T A$ is the number of shared people between language i and language j . When $i = j$, the (i, j) -th entry counts the number of people who speak language i .

$$2. \ A = \begin{pmatrix} -8 & -8 & 9 \\ 1 & 5 & -6 \\ -5 & -10 & -2 \end{pmatrix}, \quad A^{-1} = \begin{pmatrix} -0.159 & -0.241 & 0.007 \\ 0.073 & 0.139 & -0.089 \\ 0.034 & -0.091 & -0.073 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & -2 & -2 \\ -2 & -5 & -4 \\ -3 & -5 & -1 \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 0.273 & -0.145 & 0.036 \\ -0.182 & 0.164 & -0.291 \\ 0.091 & -0.382 & 0.345 \end{pmatrix}$$

$$C = \begin{pmatrix} -2 & -9 & 10 \\ -7 & 0 & -5 \\ -9 & -10 & 0 \end{pmatrix}, \quad C^{-1} = \begin{pmatrix} -0.127 & -0.253 & 0.114 \\ 0.114 & 0.228 & -0.203 \\ 0.177 & 0.154 & -0.159 \end{pmatrix}$$

Solution 5

- Lowest mean: `long` (-122.214), highest mean: `price` (540,088.142).
Lowest variance: `waterfront` (0.007), highest variance: `price` (134,776,142,225.573).
- All of the features are positively correlated with `price`:

feature	corr with <code>price</code>
<code>long</code>	0.021626
<code>condition</code>	0.036362
<code>yr_built</code>	0.054012
<code>sqft_lot15</code>	0.082447
<code>sqft_lot</code>	0.089661
<code>yr_renovated</code>	0.126434
<code>floors</code>	0.256794
<code>waterfront</code>	0.266369
<code>lat</code>	0.307003
<code>bedrooms</code>	0.308350
<code>sqft_basement</code>	0.323816
<code>view</code>	0.397293
<code>bathrooms</code>	0.525138
<code>sqft_living15</code>	0.585379
<code>sqft_above</code>	0.605567
<code>grade</code>	0.667434
<code>sqft_living</code>	0.702035

The feature with the highest correlation is `sqft_living`.

- No